ITA0608-MACHINE LEARNING

LAB PROGRAMS

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EXP NO:2 Candidate-Elimination Algorithm

AIM:

To implement and demonstrate the Candidate -Elimination algorithm for finding the most specific hypothesis from the given training data

PROCEDURE:

1. Define the training dataset
2. Initialize the hypothesis with the most specific values (0).
3. Consider only the positive training examples
4. Update the hypothesis by replacing mismatched attributes with ?
5. Display the final hypothesis

PROGRAM:

```
def find_s(training_data):
    hypothesis = None
    for sample in training_data:
        attributes = sample[:-1]
        target = sample[-1]

        if target == "Yes":
            if hypothesis is None:
                hypothesis = attributes.copy()
            else:
                for i in range(len(hypothesis)):
                    if hypothesis[i] != attributes[i]:
                        hypothesis[i] = "?"
    return hypothesis
training_data = [
    ["Sunny", "Warm", "Normal", "Strong", "Warm", "Same", "Yes"],
    ["Sunny", "Warm", "High", "Strong", "Warm", "Same", "Yes"],
    ["Rainy", "Cold", "High", "Strong", "Warm", "Change", "No"],
    ["Sunny", "Warm", "High", "Strong", "Cool", "Change", "Yes"]
]
hypothesis = find_s(training_data)
print("Final Hypothesis:")
print(hypothesis)
```

OUTPUT:

```
Final Hypothesis:
['Sunny', 'Warm', '?', 'Strong', '?', '?']
```

RESULT :

Thus ,the Candidate-Elimination algorithm was implemented successfully ,and the most specific hypothesis was obtained .